

Learning Event-Aligned Spline Actions for Language-Guided Robot Manipulation

Author names removed for blind review.

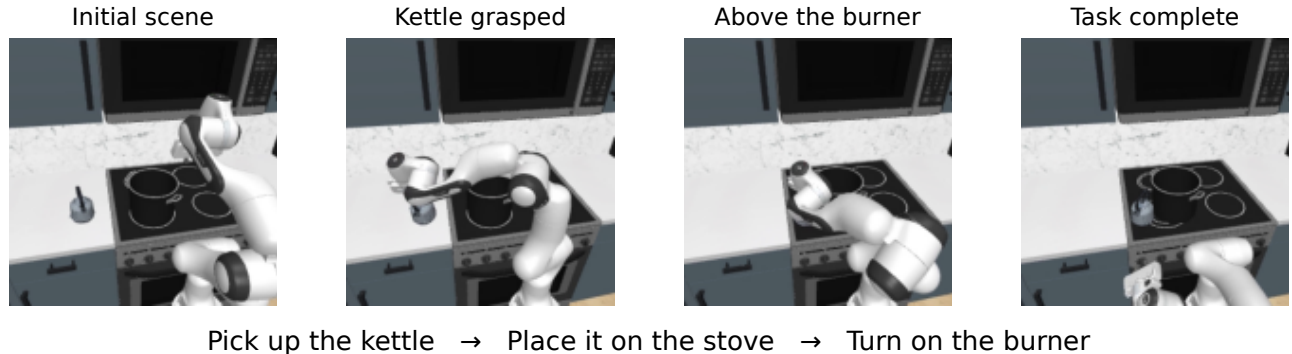


Fig. 1. A RoboCasa365 policy executes a kettle task through short language instructions. Each instruction conditions several event-aligned spline segments, each with its own predicted duration. The representation exposes two interfaces: which subgoal to execute and how to time its motion. Frames show one successful rollout; text below abbreviates the program, not the active instruction at each displayed frame.

Abstract—Long-horizon vision-language-action policies must decide both which subgoal to execute and how long to continue each predicted motion. Whole-task captions and fixed action chunks provide neither decision explicitly. We introduce event-aligned spline action programs, an interface in which ordered language clauses select the current subgoal and each policy query predicts a short command path separately from its duration. In a controlled study crossing two action heads and two label types on LIBERO-Long, training with clauses improves success under an identical clause schedule by 34.3 percentage points for a waypoint head and 43.7 points for the spline head. The resulting 9.3-point interaction is positive for all three training seeds. Reversing the clause order changes the first completed subgoal, while semantic phase labels improve aggregate success on two longer RoboCasa365 tasks in two independent training runs. The same shape-duration representation also exposes a physical clock. Retiming a frozen spline policy raises mean CALVIN chain length from 1.31 to 1.69 across three seeds at $1.42\times$ realized pace, and replanning from predicted duration preserves success on two LIBERO suites while reducing policy queries by $2.2\text{--}2.7\times$. The results show how one compact action interface can support explicit semantic and temporal control without modifying the vision-language backbone.

I. INTRODUCTION

Long-horizon manipulation requires a robot to coordinate semantic progress with continuous control. A vision-language-action (VLA) policy is commonly trained from demonstrations carrying one instruction and predicts a fixed number of future actions [1], [2], [3]. The policy must infer which part of the instruction remains active, while the fixed chunk determines how long the predicted motion lasts. As tasks acquire more subgoals, these implicit choices make it difficult to change a requested subgoal during execution or adapt the same motion to a different pace.

We study an action interface that makes both choices

explicit. An ordered language program supplies the current subgoal, and an event-aligned spline represents the motion issued under that subgoal. Each policy query predicts eight cubic B-spline control points, a gripper curve, and a duration. Training segments end at a gripper transition, a sustained pause, or a duration cap. The active clause conditions the same policy until the language program advances (Fig. 1); several predicted segments can execute one clause. Separating the command path from its duration also lets the executor change pace, sampling density, and query time after training.

We ask two questions. First, does an event-aligned spline head make piecewise language supervision more effective? We compare spline and waypoint heads using identical physical demonstrations, matched fine-tuning budgets, and the same clause schedule at test time. Second, what useful control does the learned duration provide? Both spline and waypoint paths can be resampled, so the timing study includes an interpolation baseline and separately tests the information carried by duration.

Our primary language experiment crosses action head, training labels, and inference interface. On two LIBERO-Long tasks, spline policies trained with clauses reach 55.3% success under scheduled clauses, compared with 41.3% for waypoint policies trained with clauses. Relative to whole-caption training under that *same schedule*, the spline head gains an additional 9.3 percentage points, positive in each of three seeds. The full factorial measures the trade-off between scheduled and static instructions. Reversing clauses provides a behavioral check, and official semantic phases reproduce the aggregate benefit of clause training on longer RoboCasa365 programs.

On the official CALVIN chain benchmark, changing only the spline decoder improves all three trained policies and

raises mean completed tasks from 1.31 to 1.69. The realized execution pace increases by $1.42\times$. On LIBERO, predicted duration also schedules replanning $2.2\text{--}2.7\times$ less often without reducing success. Language adverbs and changes in control rate provide complementary tests of the same physical clock.

The paper makes two contributions:

- An event-aligned spline VLA interface and a controlled study showing that it benefits more from executable language supervision than a waypoint head, supported by order interventions and RoboCasa phase supervision.
- A shape–duration decoder that exposes the physical clock of each predicted segment, evaluated through retiming, waypoint interpolation, predicted query times, language commands for pace, and changes in control rate.

II. RELATED WORK

Structured action representations. ACT and Diffusion Policy predict action chunks to model temporally extended behavior [4], [5]. VLAs combine such action prediction with pretrained visual and language representations [2], [3]; FAST compresses action sequences with frequency-domain tokens [6]. Several recent methods use splines. BEAST encodes action sequences as B-spline tokens [7]. B-spline Policy predicts continuous curves and accelerates execution by temporal rescaling [8]. Spline Policy studies continuous outputs across policy families and their use in resampling, editing, and downstream control [9]. Our contribution concerns event-duration supervision and the interaction between a structured action head and executable language. Continuous trajectories and temporal resampling are shared with this prior work.

Language and long-horizon execution. Large VLAs combine heterogeneous training data and semantic prediction to execute extended tasks [10]. Long-VLA uses phase-aware inputs [11], while tool-aligned VLAs separate high-level reasoning from policy invocation [12]. We instead study a single policy conditioned on successive clauses, with the physical training trajectories held fixed. Counterfactual instruction studies expose cases where visual or positional shortcuts dominate language [13], [14]. They motivate testing the requested behavior directly. Our order intervention therefore measures the first completed subgoal separately from final task success; changing the reward under an incompatible prompt alone would not establish successful steering.

Execution timing. PACE selects action-chunk execution horizons from low-speed transitions at inference [15]. Inference-time policy steering uses human interaction to guide generated behavior [16]. Here, demonstration events supervise segment duration, and a decoder controls its execution time. The clause scheduler and motion clock act at different scales: one chooses the active subgoal; the other determines how a predicted segment is sampled.

III. EVENT-ALIGNED SPLINE ACTIONS

Let o_t denote images, s_t robot state, and ℓ_t the active instruction. A conventional chunking VLA predicts H future actions from (o_t, s_t, ℓ_t) . We retain its vision–language backbone and flow-matching action expert, but change the prediction targets and decoder (Fig. 2).

A. A Short Motion Program

For a window of T pose-delta commands $a_0^{\text{pose}}, \dots, a_{T-1}^{\text{pose}}$, form the cumulative command path

$$p_0 = 0, \quad p_k = \sum_{j=0}^{k-1} a_j^{\text{pose}}. \quad (1)$$

This path is expressed in the controller’s command coordinates. It is not a reconstruction of the measured end-effector pose, and summing rotational command components is not an exact composition on $SO(3)$. We approximate it with a clamped, uniform cubic B-spline,

$$p(u) = \sum_{i=0}^{n-1} B_{i,3}(u)c_i, \quad u \in [0, 1], \quad n = 8. \quad (2)$$

The training fit pins $c_0 = 0$ and $c_{n-1} = p_T$. The remaining controls minimize

$$\sum_{k=0}^T \|p(k/T) - p_k\|_2^2. \quad (3)$$

We precompute the least-squares operators for each allowed duration. The gripper command is fitted directly, without cumulative summation, and thresholded by sign during decoding. In RoboCasa’s mobile-manipulator action layout, base and mode commands occupy additional fitted channels rather than pose-delta channels.

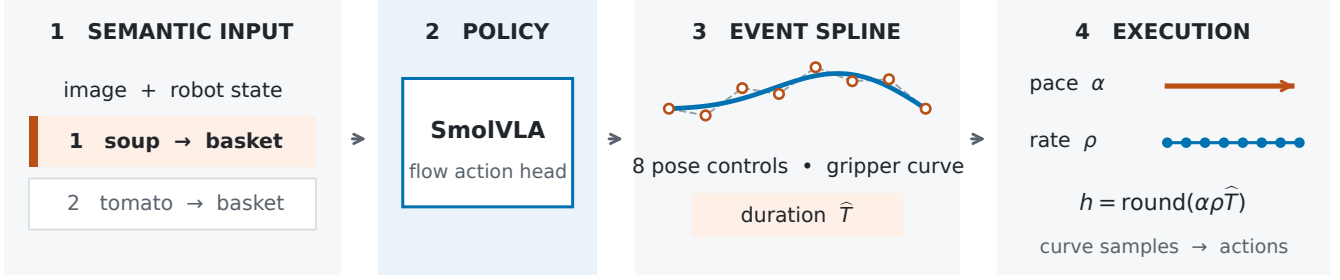
Each output token contains one pose control point, one gripper control value, and a replicated duration target $\log T$. The action expert therefore emits eight tokens rather than the waypoint head’s 50. We standardize targets per token and train with the same conditional flow-matching objective [3], [17]. At inference, we average the predicted log durations, $\bar{z} = \frac{1}{n} \sum_i \widehat{\log T}_i$, and decode

$$\hat{T} = \text{clip}(\text{round}(e^{\bar{z}}), T_{\min}, T_{\max}). \quad (4)$$

We reset the predicted starting control point to zero. The predicted endpoint remains learned; endpoint-preserving resampling preserves that prediction, not necessarily the demonstrated endpoint or the robot’s realized path.

B. Event-Aligned Supervision

Starting at each training frame, we select the first boundary after $T_{\min} = 8$ commands. A boundary is a gripper-command transition, two consecutive pose-command norms below 0.15 times the window median, or episode padding. If none occurs, the window ends at $T_{\max} = 24$ commands (16 for CALVIN). A boundary at index k gives $T = k$: commands $0, \dots, k-1$ enter the fit, while the boundary command belongs to the next segment. Thus duration predicts an action event or cap, not a semantic subgoal label.



A clause selects the subgoal; several event-aligned segments execute it.

Fig. 2. The active clause conditions an unchanged VLA backbone. Each query produces a short spline segment and its duration. The decoder chooses the number of issued actions using duration scale α and control-rate ratio ρ . Clause switching occurs on a separate, slower schedule; it is held identical across heads in the primary language comparison.

This distinction matters for language supervision. Gripper and pause events provide short motion boundaries, but a task clause may require several grasps, transports, or contact corrections. Event-aligned targets do not impose a one-to-one correspondence between language clauses and spline segments.

C. Executable Clauses

A program is an ordered list $\mathcal{L} = (\ell^1, \dots, \ell^M)$. At any time, the active clause replaces the whole-task caption in the policy input. A scheduler advances the clause pointer and clears queued actions generated under the previous instruction. There is no additional high-level model.

For the two-subgoal LIBERO study, we label each demonstration with two clauses. The split is the first gripper release following at least 48 consecutive closed commands; the release is assigned to the first clause, and the next observation-action pair begins the second. The two training views have identical images, states, actions, and episode membership. Only language metadata changes.

The main comparison uses an outcome-independent, fixed clause clock for both heads. The switch occurs at the median labeled transition in the training demonstrations: step 139 for the soup/tomato task and 107 for the two-mug task. A second condition switches at an executed close-to-open gripper command. The order diagnostic instead uses the official subgoal predicate to advance the clause; this privileged scheduler isolates the order request from completion-detection errors and is not a deployable result.

D. Controlling the Physical Clock

Let f_0 be the action rate used to interpret the training commands, f the execution rate, and $\rho = f/f_0$. A duration multiplier α yields

$$\begin{aligned} h &= \max\{2, \text{round}(\alpha \rho \hat{T})\}, \\ a_j^{\text{pose}} &= p((j+1)/h) - p(j/h). \end{aligned} \quad (5)$$

Ignoring rounding and actuator clipping, execution time is $h/f \approx \alpha \hat{T}/f_0$. Changing α therefore changes pace; changing ρ at fixed α changes sampling density while preserving duration. The underlying predicted curve is unchanged. The

grripper curve is sampled on the corresponding grid and sign-thresholded. As with the original controller, issued pose commands are bounded.

This factorization guarantees a shared predicted endpoint across sampling grids before clipping. It does not guarantee unchanged closed-loop outcomes, tracking at arbitrary speeds, or derivative continuity across replans. We use ordinary receding-horizon execution: issue up to K actions from a segment, observe the robot again, and replan. A segment shorter than K drains the queue earlier.

Waypoint interpolation baseline. The same timing operation can be applied to a waypoint prediction. Denormalize its per-step deltas, accumulate them into a path q_k , linearly interpolate that path at $h+1$ equally spaced phases, and difference successive samples. The gripper uses nearest-step sampling and sign thresholding. Finally, renormalize if the policy's saved postprocessor expects normalized actions. Interpolating normalized deltas directly is incorrect: adding a per-step mean after changing the number of commands changes the total displacement. This baseline needs no re-training and is distinct from repeating unchanged waypoints at another rate.

IV. EXPERIMENTS

We organize the evaluation around two questions: whether event-aligned spline actions improve execution from piecewise language, and whether their learned duration provides useful temporal control. Broad benchmark success provides context for both interventions.

A. Setup

Policies and data. All heads use the same 450M-parameter SmolVLA backbone [3]. Waypoint A predicts 50 action tokens; fixed-window spline B uses eight control points and a 20-command target; event spline C uses eight control points and predicts duration. The broad LIBERO study trains each for 100,000 updates with batch size 32 on the same 40-task dataset, with three training seeds. The clause study begins from the corresponding 100,000-update A and C checkpoints, crossing whole-task and clause-aligned training labels. It uses 7,500 updates, batch size 32, and the

TABLE I

LIBERO-LONG T0/T4 SUCCESS (%). EACH CELL CONTAINS 300 ROLLOUTS: 50 STATES/TASK $\times$ TWO TASKS $\times$ THREE TRAINING SEEDS. COLUMNS CHANGE INFERENCE; ROWS CHANGE TRAINING LABELS.

Head	Training labels	Inference instruction	
		Static whole	Clauses
Waypoint	Whole-task	49.0	7.0
Waypoint	Clauses	39.7	41.3
Event spline	Whole-task	57.0	11.7
Event spline	Clauses	33.0	55.3

same physical episodes for every treatment. Evaluation uses the final checkpoint rather than selecting the best checkpoint on test outcomes.

Evaluation. LIBERO [18] supplies four ten-task suites. The language experiment uses two multi-object LIBERO-Long tasks: soup and tomato into a basket (t0), and two mugs onto different plates (t4). Each condition evaluates the same 50 simulator states per task and three training seeds. The primary comparison sets $K = 10$ and the same fixed clause schedule for both heads. Exact replays verify initial observations, simulator states, prompts, action traces, and outcomes. Replay repeats are checks, not additional independent trials.

RoboCasa365 [19] tests KettleBoiling and RinseSinkBasin on a targeted scene split, with official full-task success predicates. CALVIN [20] uses the official list of five-instruction chains and reports mean consecutive tasks completed. The existing timing study evaluates 1,000 chains per checkpoint seed. Success percentages always refer to complete tasks unless a stage metric is named explicitly.

Uncertainty. For the primary language interaction, we report the effect for each of three trained policies and the t interval over their training seeds. A hierarchical bootstrap resamples seeds and paired task/state outcomes as a sensitivity analysis. For binary paired interventions we use exact McNemar tests; the reported tests of RoboCasa phase labels use Fisher’s exact test. CALVIN’s historical intervals resample paired chains for the evaluated checkpoints; they are not population intervals over possible training runs.

B. Question 1: Does the Action Head Affect Piecewise Language Execution?

The language dataset contains 33 demonstrations of t0 and 38 of t4: 71 episodes and 19,378 frames. Table I crosses the training labels with the inference interface. The scheduled columns use the same clauses and transition times regardless of training treatment. This holds the requested language program fixed while testing whether the policy learned to execute it.

Under scheduled clauses, success rises from 7.0% to 41.3% for the waypoint head and from 11.7% to 55.3% for the event spline. The label effects are +31, +30, and +42 points for A, and +41, +42, and +48 for C (Fig. 3). Define

TABLE II

ORDER INTERVENTION. FIRST-OBJECT FLIPS ARE MEASURED ON T4 (50 PAIRED STATES); FINAL TASK COUNTS POOL T0/T4 (100 STATES). ALL POLICIES HERE ARE CLAUSE-TRAINED.

Head	First-object flips		Final success	
	Requested	Opposite	Normal	Reverse
Waypoint	10/50	0/50	49/100	16/100
Event spline	10/50	0/50	55/100	21/100

the head interaction at the fixed schedule as

$$\Delta = (S_{C,\text{clause}} - S_{C,\text{whole}}) - (S_{A,\text{clause}} - S_{A,\text{whole}}), \quad (6)$$

where the second subscript denotes *training labels*. Its mean is +9.33 points, with a 95% t interval of [1.74, 16.92] over training seeds and a hierarchical interval of [0.67, 18.0]. The advantage is therefore not limited to a single training run. It concerns the event spline head as a whole; this experiment does not separately identify the contributions of control-point count, event boundaries, and duration supervision.

The static columns provide an important control. Whole-caption-trained policies remain competent when given their native whole-caption input. Clause training does not improve this interface: it reduces static-prompt success by 9.3 points for A and 24.0 for C. The useful result is improved *addressability through a clause program*, not a universal increase in canonical-task success. With the spline head, programmed success reaches 55.3%, close descriptively to the whole-caption model’s 57.0%, while allowing the subgoal instruction to change during execution.

Switching without a fixed demonstration clock. Advancing the spline’s clauses after its gripper opens yields 164/300 successes for clause training and 43/300 for whole-caption training: a 40.3-point label effect. This retains the direction and scale of the result obtained with a fixed clause schedule. It uses an action-derived switch, not the scalar duration prediction or an oracle goal predicate.

C. Changing the Requested Subgoal Order

We reverse the two clauses and score which official goal predicate becomes true first. This diagnostic uses a privileged predicate to advance to the second clause. On t4, both heads exhibit 10/50 paired states in which the first completed object changes in the requested direction, with no changes in the opposite direction (two-sided sign test $p = .00195$). Of 12 states with a unique first goal under both orders, 10 switch as requested.

The intervention changes behavior, but final conjunction success falls under reversed order (Table II). The policy can respond to the first requested subgoal without reliably preserving it while completing the second. This separates first-subgoal steering from compositional generalization, and does not imply a spline-specific order advantage.

D. Semantic Phases on RoboCasa365

RoboCasa tests whether language alignment helps longer sequences involving grasping, transport, and fixture actua-

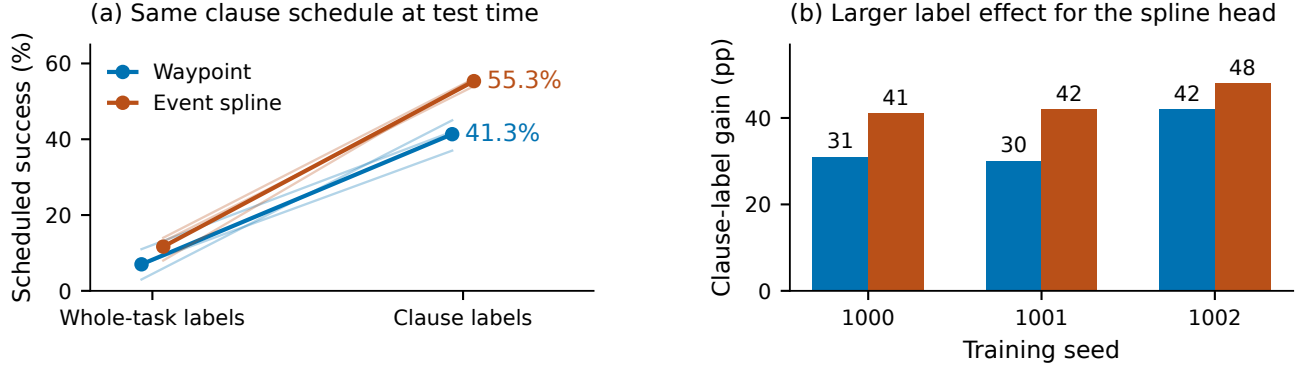


Fig. 3. Clause-aligned training improves execution under the same fixed clause schedule. (a) Absolute success on t0/t4; faint lines show individual training seeds and thick lines show their means. (b) The label effect for each seed. The difference between the spline and waypoint effects is +10, +12, and +6 percentage points.

TABLE III

ROBOCASA FULL-TASK SUCCESS COUNTS UNDER THE SAME FIXED PHASE SCHEDULE. n GIVES THE NUMBER OF TRIALS PER ROW AND LABEL TREATMENT. COMBINED ROWS SUM THE TWO TASKS.

Seed	Task	n	Whole labels	Phase labels
1000	Kettle	50	2	6
	Rinse	50	2	10
	Combined	100	4	16
1001	Kettle	20	2	1
	Rinse	20	1	8
	Combined	40	3	9

tion. We use the same event-spline architecture and relational identical demonstrations with official semantic phases. KettleBoiling has 501 demonstrations and 228,349 frames; RinseSinkBasin has 509 demonstrations and 211,036 frames. Within each seed, whole-caption and phase-label policies share initialization and receive 30,000 updates at batch size 32. Both are evaluated under the same outcome-independent phase schedule: Kettle switches at steps 128 and 242; Rinse switches at 162. These are demonstration-derived constants, not test-time success signals.

The two-task aggregate improves by 12 points on seed 1000 and 15 on seed 1001 (Table III). The first aggregate has Fisher $p = .00808$. Rinse improves on both seeds; Kettle’s improvement does not replicate in the smaller evaluation of the second training seed. The result compares labels on two tasks in selected scenes; it is not an estimate over all 365 tasks or a waypoint–spline interaction on RoboCasa.

Stage measurements help locate the improvement. On seed 1000, phase labels increase Kettle grasp attainment from 24/50 to 39/50, and Rinse attainment of water-on with at least one washed region from 7/50 to 22/50. The latter improves from 3/20 to 12/20 on seed 1001. These intermediate counts are not substituted for full-task success.

The teaser follows one successful phase-trained Kettle rollout from a paired evaluation. Its whole-caption-trained counterpart begins from the identical observation and also

TABLE IV

CALVIN MEAN CONSECUTIVE TASKS COMPLETED. RETIMING USES $\alpha = 0.6$, WITH ACTUATOR CLIPPING AND NO FEASIBILITY STRETCH. EACH SEED HAS 1,000 CHAINS.

Policy / decoder	1000	1001	1002	Mean
Waypoint / native	1.510	1.764	1.416	1.563
Spline / native	1.422	1.239	1.273	1.311
Spline / retimed	1.734	1.579	1.759	1.691

grasps and reaches the stove. Only the phase-trained policy completes burner actuation: success at 516 steps versus failure at the 1,500-step limit. This paired example localizes the difference to the final stage.

E. Question 2: What Does the Learned Physical Clock Enable?

The timing experiments change the decoder of an already trained policy. CALVIN is a useful test because its human demonstrations include pauses and variable pace. The event head uses eight control points, $T_{\min} = 8$, and $T_{\max} = 16$ at the policy’s 10-Hz action rate. Each command is held for three native simulator steps, preserving the replay-validated action convention. Every subtask receives the official 360-simulator-step budget.

Uniform retiming improves the spline for every training seed (Table IV), with a mean gain of 0.379 completed tasks over paired chains (95% CI [0.318, 0.441]). Realized pace is $1.42\times$ the original spline, rather than the nominal $1/0.6 = 1.67\times$, because rounding, replanning, and changed behavior also affect completion time. The retimed spline exceeds the native waypoint by 0.127 tasks on average (95% CI [0.063, 0.191] over paired chains), while first-task success is similar, 69.9% versus 69.4%. Figure 4 shows that the difference is concentrated in sustaining longer chains. This historical comparison uses a native waypoint decoder; the interpolation control below addresses that limitation directly.

Separating interpolation from learned representation. Native waypoint replay does not answer whether a spline is

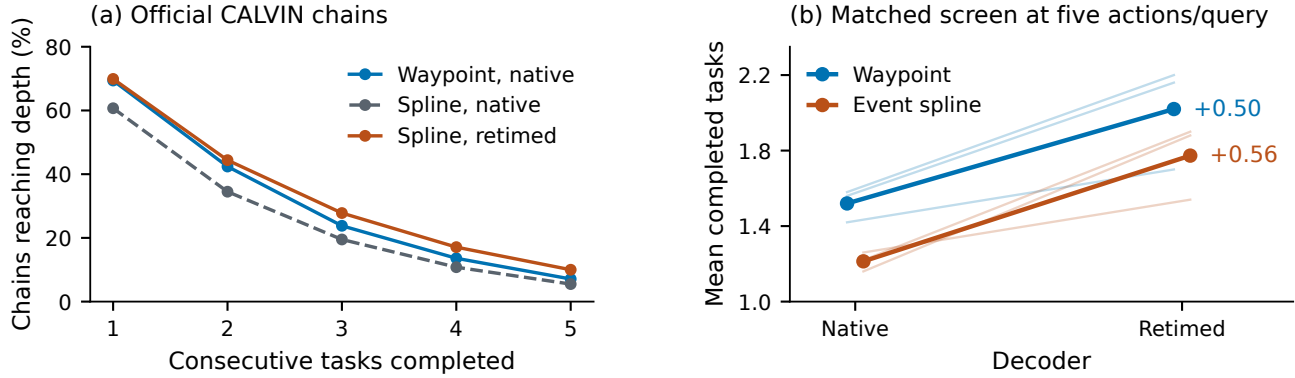


Fig. 4. CALVIN timing results. (a) Historical study with 1,000 official chains per training seed. The waypoint comparator uses its native decoder. (b) Fresh screen with 50 common chains per seed and five actions per policy query. Thin lines show individual training seeds; thick lines show means. Both representations benefit from retiming.

a better representation for retiming. We therefore specify a four-arm comparison: native and linearly retimed waypoints, and native and uniformly retimed event splines. Both retimed arms use $\alpha = 0.6$, interpolation or decoding in raw command units, the same bounds, and no feasibility stretch. All four issue five actions per neural query, so even the shortest retimed spline segment fits the common cadence. They use the same official chains, noise seeds for every chain and subtask, controller rate, and time budgets. Tests of identity, normalization with a nonzero mean, and endpoints precede closed-loop evaluation.

The matched screen confirms that interpolation is a strong baseline (Fig. 4b). Linear waypoint retiming raises mean chain length from 1.52 to 2.02; spline retiming raises it from 1.21 to 1.77. The corresponding gains are 0.50 and 0.56 tasks, giving an interaction of 0.06 (95% t interval over training seeds $[-0.87, 0.99]$). Thus temporal resampling itself is not unique to splines. The distinction is that the event spline predicts a duration that can decide *which* segments to retime and *when* to query again. The screen uses 50 official chains for each of three training seeds.

Using duration as a control signal. Retiming is only one use of the physical clock. Table V collects three interventions that ask whether the predicted timing remains operational under changes to policy queries, language, and controller rate. These are targeted tests rather than additional headline tasks.

The replanning result tests whether $\hat{T}$ predicts a useful event time rather than merely serving as a decoder length. Replanning four actions before that time preserves the reference success on LIBERO Object and Long while using $2.68\times$ and $2.24\times$ fewer neural queries. The margin matters: waiting until $\hat{T}$ reduces success to 84% and 55%, while margins of two and four actions improve monotonically. Duration can also select which motions to compress. On LIBERO Object, applying $\alpha = 0.6$ only when $\hat{T} > 20$ yields 95% success in 122.7 steps, compared with 93% in 145.3 steps at the original pace and 73% in 109.6 steps when every

segment is compressed ($n = 100$ each).

Language can also act on timing. We label CALVIN segments with kinematic adverbs derived from demonstration speed and train without changing their actions. At test time, “quickly” and “slowly” change issued pose-command magnitude by approximately +16% and −16% relative to the same instruction without an adverb. The faster condition improves chain length from 1.382 to 1.553 over 600 chains ($p = .0175$, Bonferroni-corrected $p = .035$); the neutral prefix control gains only 0.025 tasks. This measures commanded motion and task completion, not guaranteed end-effector speed under arbitrary actuator limits.

Finally, changing the control rate while increasing the number of samples tests the distinction between duration and sampling density in Eq. 5. On LIBERO Spatial t5, the event spline improves under twice the rate in two independently specified horizon protocols: 38% to 68% ($p = .00408$) and 34% to 72% ($p = .000311$), each on 50 paired states. The effect is task dependent: broad suite averages do not improve universally, so we use it as evidence that the curve can be resampled for controller adaptation rather than as a general success claim.

F. General Task Performance

Table VI gives the four-suite context for these focused studies. Across three seeds, the event head averages 80.7% against 82.1% for waypoints, a 1.4-point difference at the native rate. The two-task language factorial is evaluated separately and is not folded into these suite averages.

G. Real Robot Evaluation

Reserved for hardware results; no physical trials are claimed in this draft. The planned setup uses one 7-DoF arm with a linear base, a gripper, and handheld teleoperation. The base remains fixed unless reach requires it. Existing printed fixtures support a short drawer program: open, insert an object, then close or leave open. The two final clauses share the same prefix, making the requested behavioral difference

TABLE V
THREE USES OF THE LEARNED PHYSICAL CLOCK. POLICY WEIGHTS REMAIN FIXED IN EVERY INTERVENTION.

Control	Intervention	Outcome
Query time	Replan at $\hat{T} - 4$ on LIBERO Object / Long	$2.68 \times / 2.24 \times$ fewer queries at matched success
Language pace	Add “quickly” on CALVIN	Commands +16%; chain length $1.382 \rightarrow 1.553$
Sampling rate	Execute at $2 \times$ on LIBERO Spatial t5	Success $38 \rightarrow 68\%$ and $34 \rightarrow 72\%$

TABLE VI
BROAD LIBERO SUCCESS (%). THREE SEEDS, 1,200 ROLLOUTS/HEAD ACROSS FOUR SUITES. THE FINAL COLUMN IS MEAN $\pm$ SEED STANDARD DEVIATION OF SUITE-AVERAGED SUCCESS.

Head	Object	Spatial	Goal	Long	Average
Waypoint	94.7	84.3	86.0	63.3	82.1 ± 1.8
Fixed spline	96.3	77.7	84.0	65.3	80.8 ± 2.5
Event spline	92.3	77.3	89.3	63.7	80.7 ± 1.3

directly visible. A second feasible object-ordering task can test a changed sequence if demonstration collection permits.

The final section will report the collected demonstrations, train/test program split, matched policy comparisons, and complete-task counts on frozen initial configurations. A paired filmstrip that changes a late clause, together with a trajectory comparison before and after that change, will test whether the policy responds at the requested stage. The section will occupy approximately one page, including the result table and images. This is a focused test of language execution, not simply a deployment photograph.

V. DISCUSSION AND CONCLUSION

Event-spline policies benefit more from clause-aligned training than waypoint policies under an identical clause schedule. This interaction repeats across three seeds. The complete interface factorial shows the resulting trade-off: adapting to scheduled clauses improves their execution but reduces performance with a static whole-task caption. Order interventions show a corresponding change in the first completed subgoal, while RoboCasa extends the label comparison to longer manipulation sequences.

Several boundaries remain. The head comparison uses one VLA backbone and two LIBERO tasks, and combines event alignment, compactness, and duration prediction. Reversed-order completion remains difficult, so the current results do not establish general recombination of learned skills. Fixed clause clocks also assume predictable phase timing; the action-release condition relaxes that assumption only for tasks with a suitable gripper event.

Separating a predicted path from its duration lets the executor retime a motion without retraining the policy. CALVIN shows improved completion and pace for each frozen spline policy. An equally retimed waypoint is a strong control, establishing that generic temporal resampling is shared by both representations. The learned event duration provides the additional interface: it selects motions for compression and predicts when a new observation is useful.

Together, the studies evaluate one action representation for selecting subgoals and controlling the timing of their motion.

REFERENCES

- [1] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, *et al.*, “OpenVLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [2] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv preprint arXiv:2410.24164*, 2024.
- [3] M. Shukor, D. Aubakirova, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti, S. Alibert, M. Cord, T. Wolf, and R. Cadene, “SmolVLA: A vision-language-action model for affordable and efficient robotics,” *arXiv preprint arXiv:2506.01844*, 2025.
- [4] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” in *Robotics: Science and Systems*, 2023.
- [5] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Robotics: Science and Systems*, 2023.
- [6] K. Pertsch, K. Stachowicz, B. Ichter, D. Driess, S. Nair, Q. Vuong, O. Mees, C. Finn, and S. Levine, “FAST: Efficient action tokenization for vision-language-action models,” *arXiv preprint arXiv:2501.09747*, 2025.
- [7] H. Zhou, W. Liao, X. Huang, Y. Tang, F. Otto, X. Jia, X. Jiang, S. Hilber, G. Li, Q. Wang, *et al.*, “BEAST: Efficient tokenization of B-splines encoded action sequences for imitation learning,” in *Advances in Neural Information Processing Systems*, 2025.
- [8] X. Han, H. Xiong, H. Chen, C. Liu, A. Torralba, Y. Zhu, and Y. Du, “B-spline policy: Accelerating manipulation policies via B-spline action representations,” *arXiv preprint arXiv:2607.09648*, 2026.
- [9] M. Tian, Y. Li, S. Liu, A. Ijspeert, and S. Calinon, “Spline policy: A structured representation for robot policies,” *arXiv preprint arXiv:2606.07386*, 2026.
- [10] Physical Intelligence, K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, *et al.*, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” *arXiv preprint arXiv:2504.16054*, 2025.
- [11] Y. Fan, P. Ding, S. Bai, X. Tong, Y. Zhu, H. Lu, F. Dai, W. Zhao, Y. Liu, *et al.*, “Long-VLA: Unleashing long-horizon capability of vision language action model for robot manipulation,” in *Conference on Robot Learning*, 2025.
- [12] Z. Lei, C. Liu, Y. Xiong, M. Xiong, Y. Ding, Z. Zhang, W. Li, and S. Chen, “Towards long-horizon embodied agents with tool-aligned vision-language-action models,” *arXiv preprint arXiv:2605.13119*, 2026.

- [13] C. Glossop, W. Chen, A. Bhorkar, D. Shah, and S. Levine, "CAST: Counterfactual labels improve instruction following in vision-language-action models," *arXiv preprint arXiv:2508.13446*, 2025.
- [14] Y. Fang, Y. Feng, D. Jing, J. Liu, Y. Yang, Z. Wei, D. Szafir, and M. Ding, "When vision overrides language: Evaluating and mitigating counterfactual failures in VLAs," *arXiv preprint arXiv:2602.17659*, 2026.
- [15] J. Nie, J. Li, J. Zhang, J. Lao, C. Liu, T. Zhang, L. Lin, and S. Huang, "PACE: Phase-aware chunk execution for robot policies with action chunking," *arXiv preprint arXiv:2606.00537*, 2026.
- [16] Y. Wang, L. Wang, Y. Du, B. Sundaralingam, X. Yang, Y.-W. Chao, C. Perez-D'Arpino, D. Fox, and J. Shah, "Inference-time policy steering through human interactions," in *IEEE International Conference on Robotics and Automation*, 2025.
- [17] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *International Conference on Learning Representations*, 2023.
- [18] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "LIBERO: Benchmarking knowledge transfer for lifelong robot learning," in *Advances in Neural Information Processing Systems*, 2023.
- [19] S. Nasiriany, S. Nasiriany, A. Maddukuri, and Y. Zhu, "RoboCasa365: A large-scale simulation framework for training and benchmarking generalist robots," in *International Conference on Learning Representations*, 2026.
- [20] O. Mees, L. Hermann, E. Rosete-Beas, and W. Burgard, "CALVIN: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks," *IEEE Robotics and Automation Letters*, 2022.